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palm
PalmSource
Conference 2000

Developing for Symbol's Palm OS® Handhelds



Symbol Technologies, Inc.
Mobile Computing Systems

- Product Marketing
- Software Engineering
- Software Marketing

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Today's Agenda

- Symbol and our Products
  - Products, Markets and Directions
- How to develop for Symbol's handhelds
  - What you need to know to develop
- How to begin to work with Symbol
  - Working with Symbol as a Developer

Symbol's Core Values

- Integrated Data Capture
  - Bar Code Scanning
    - One dimensional scanning
    - Two dimensional scanning
    - Imaging
- Integrated Wireless Communications
  - Wireless Ethernet for data
  - Wireless Ethernet for voice
- Rugged Ergonomics
- Customization and Strategic Alliances
  - Tools, emulation software, printing, accessories



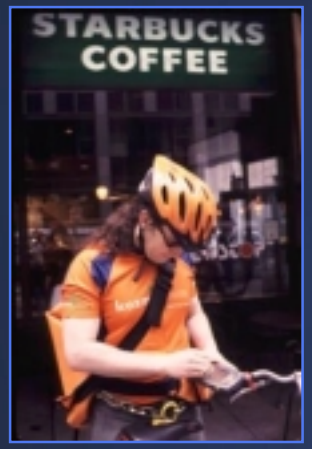


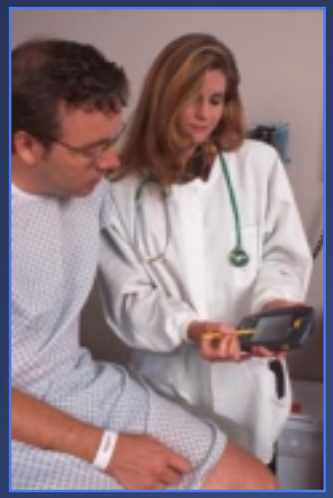
Where Is Symbol in the Palm Economy



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Markets for Symbol's Handhelds

















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Customers Deploying Symbol's Handhelds



Our Products Today









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Our Direction in the Future

- Strengthening our value to the platform
  - Data capture
    - SOAC
    - Wireless imaging
  - Wireless convergence
    - Data, voice and imaging
- Improving ergonomics
  - Size, weight, wearability
- Enhancing our features
  - Speed, display, memory, battery life



Symbol's Palm OS Engineers

Carol Sigda—Sr. Software Engineer
John Du Hart, IV—Principal Engineer
Mike Faith—Sr. Software Engineer

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SPT SDK

Palm OS® Versions

- All SPT terminals use Palm OS, version 3.5.2
- Contents of the SDK
  - Printing Capability
  - Magnetic Stripe Reader
  - Spectrum24 Wireless Radio
  - 2D Barcodes
  - WWAN

SPT SDK

- Scanning
- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio
- CDPD
- GSM



SPT SDK Contents

- SDK is the same for the SPT 1500 and SPT 17xx
- What is the same?
  - Palm OS 3.5.2
  - API has not changed
  - Scan Manager Library
  - Flash Utilities
  - ROM Builder Tools
  - Development Tools
    - CodeWarrior
    - Satellite Forms
    - gcc, etc.

SPT SDK—Enhancements

- Contents of the SDK
  - Printer API
  - Magnetic Stripe Reader API
  - S24 Wireless Radio API
  - CDPD and GSM
  - Sample Code for Scanning, Printing, MSR and S24
- Documentation
- Libraries
  - Printing
  - MSR
  - S24

SPT SDK Features

- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio
- WWAN



Why Printing?

- Add printing capability for barcode printers
- Add printing for receipt printers
- Allow users with commercial printers to print using Postscript and PCL

Printers Currently Supported

- Label and Barcode Printers
  - Comtec
  - Eltron
  - Monarch
  - O'Neil
- Commercial Printers
  - PCL
  - Postscript
- Additional Printers
  - Can add new printer support

Printer API Features

- High-level text and Graphics
- Generic write function
- Supports many printer types
- Ability to handle multiple transports
  - Serial
  - Ir

SPT SDK

- Scanning
- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio
- WWAN

MSR 3000 Features

- Attachment for the Symbol SPT 1700/1740
- Interfaces via the HotSync® port
- Powered from SPT 17xx
- Barrel connector for Charging SPT 17xx Battery
- Reads One, Two, or Three Track cards



MSR 3000 Features

- Decodes most common formats
  - American National Standard magnetic stripe encoding format (ISO-7811)
  - AAMVA recommended format (Driver's Licenses)
  - California DMV magnetic stripe encoding format
- Generic and Raw Decode modes for non-standard formats
- Implemented by MsrMgrLib.prc shared library

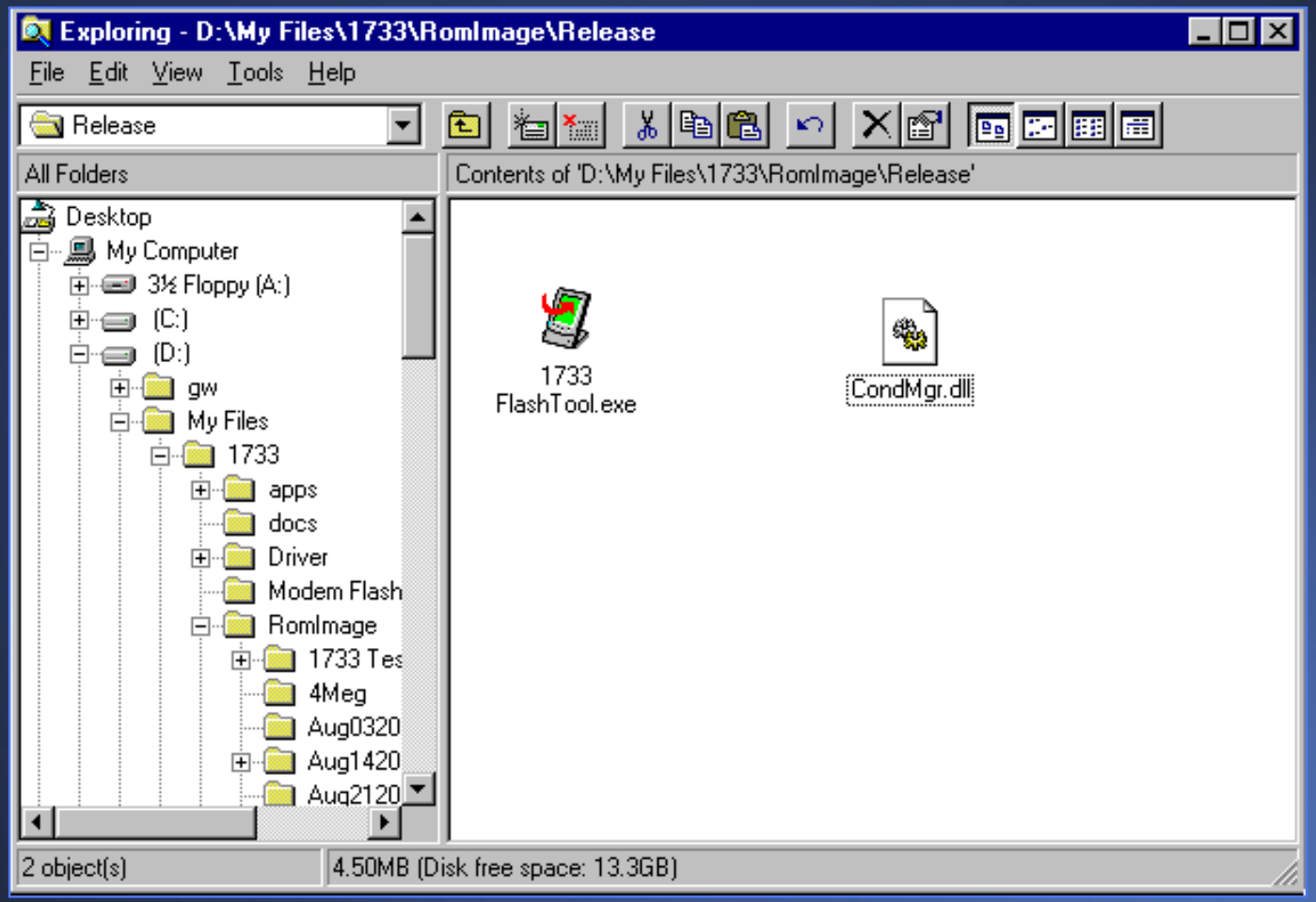
How to Add Files to the SPT Terminals

- Flash
- HotSync®
- ROM Builder
- Wireless

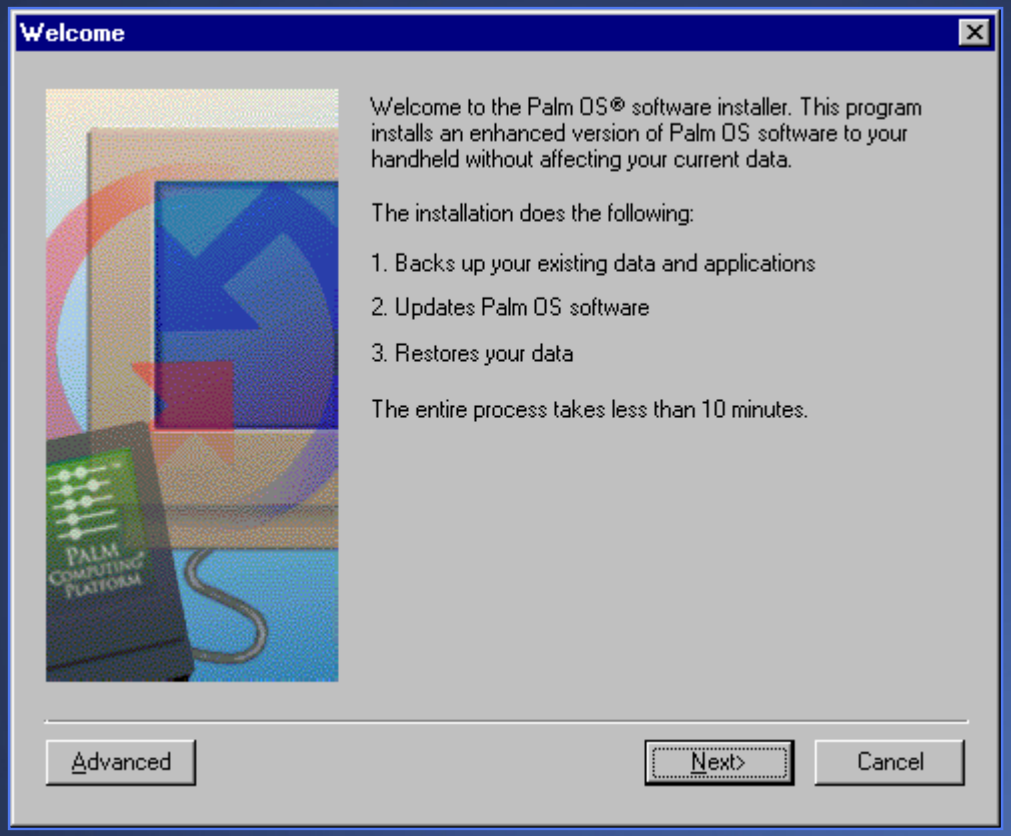
Methods of Adding Files to a SPT

- ROM Flash
  - By using FlashUtilities:
    - 1700BigRom.1D.310.exe
    - condmgr.dll
    - Instructions
- Synchronization
  - HotSync®
  - Using third-party tools
- Wireless Synchronization
  - HotSync over Spectrum24

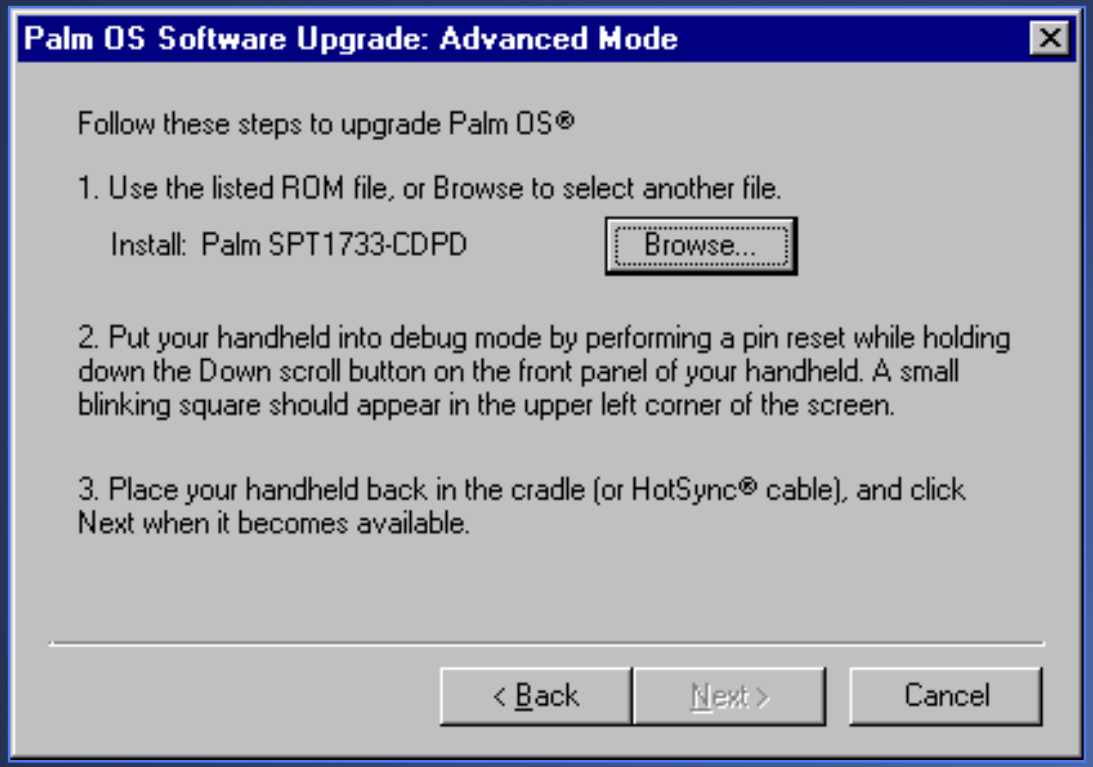
Using the Palm Flash Tool



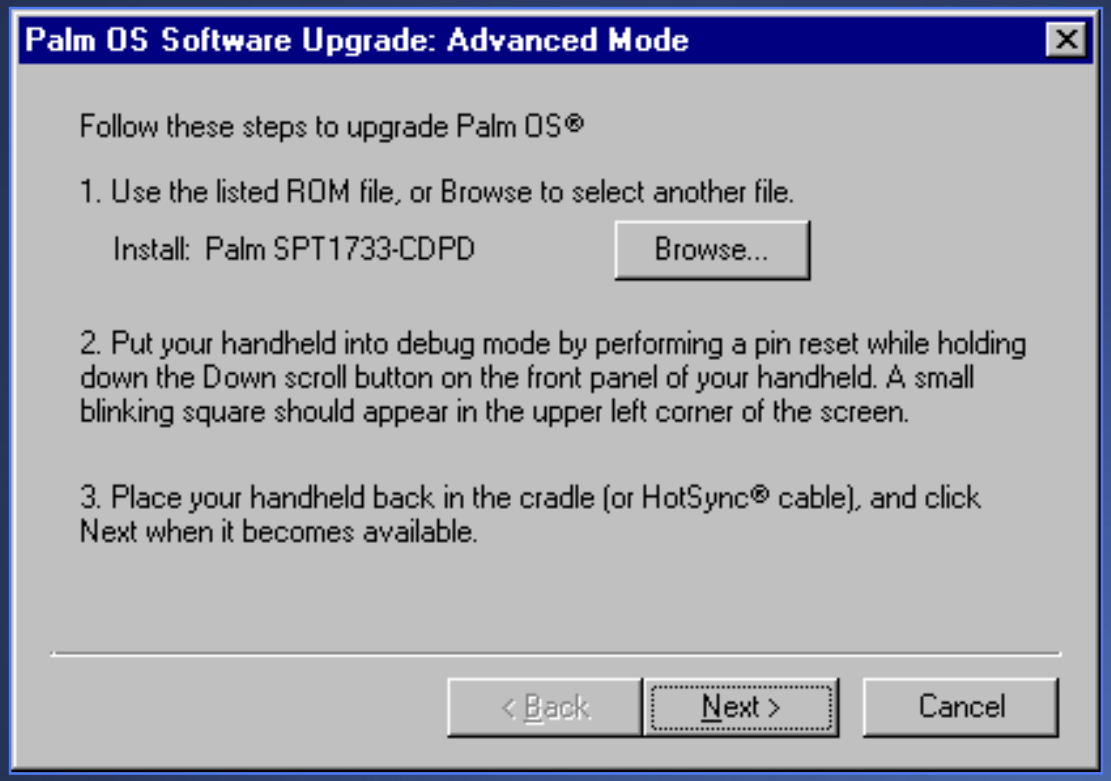
Flash Tool—Welcome



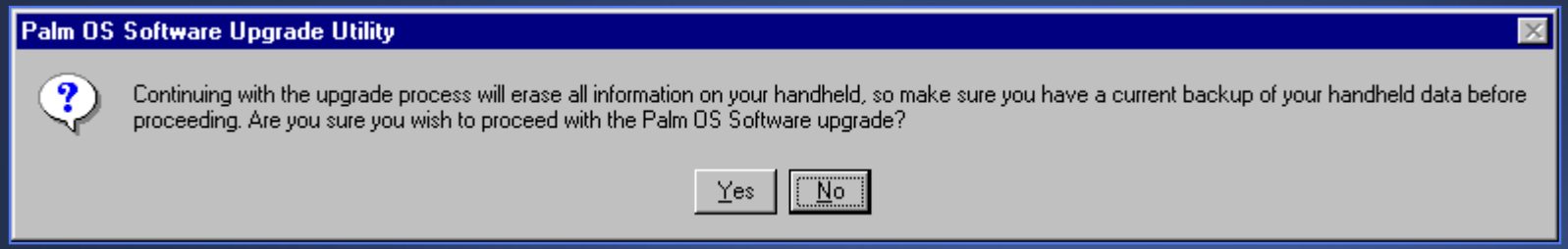
Flash Tool—Advanced



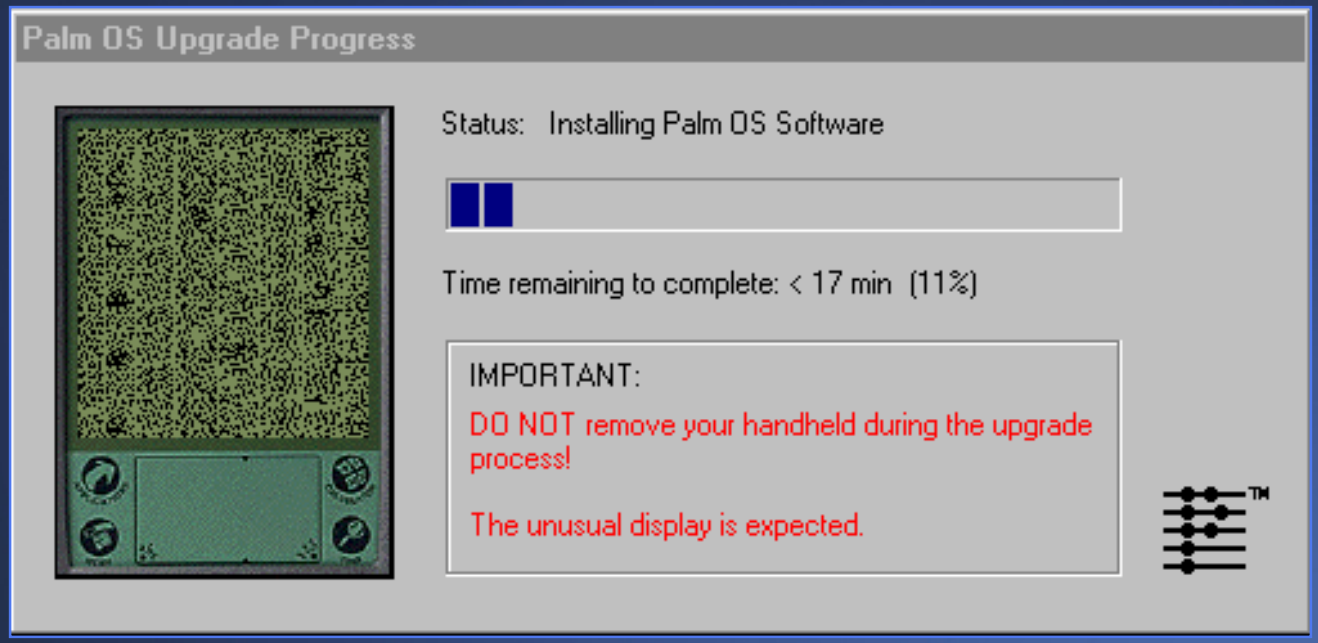
Flash Tool—Next



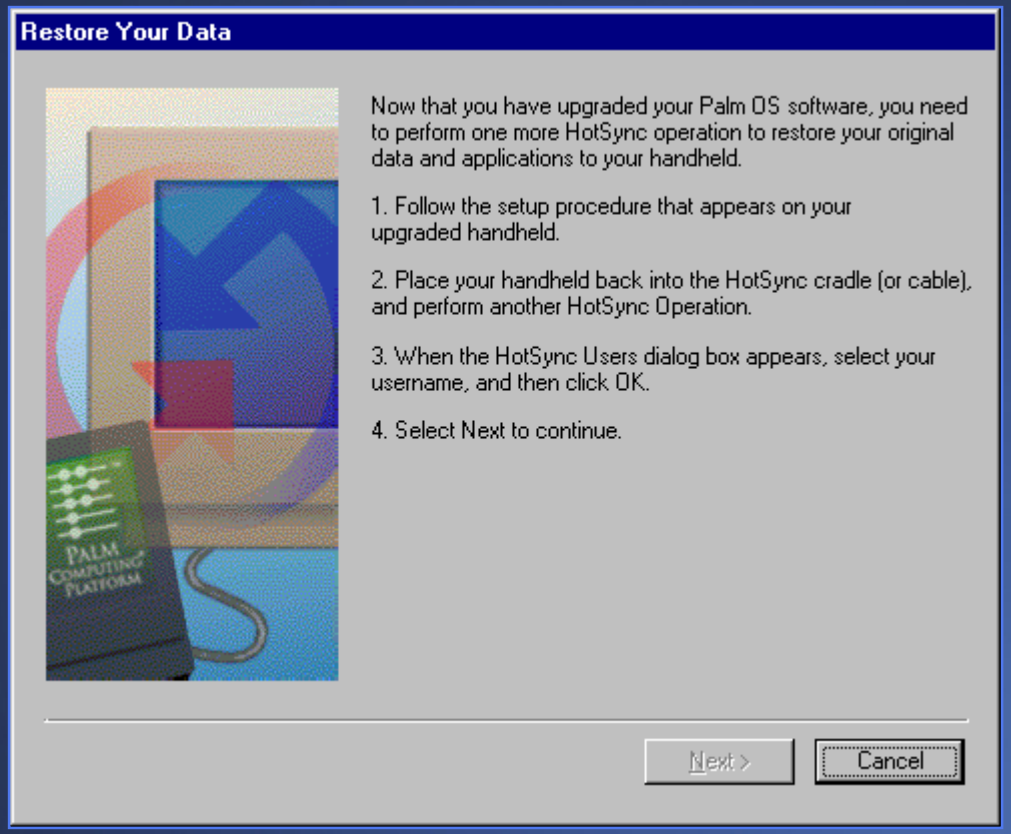
Flash Tool—Are You Sure?



Flash Tool—Progress



Flash Tool—Done



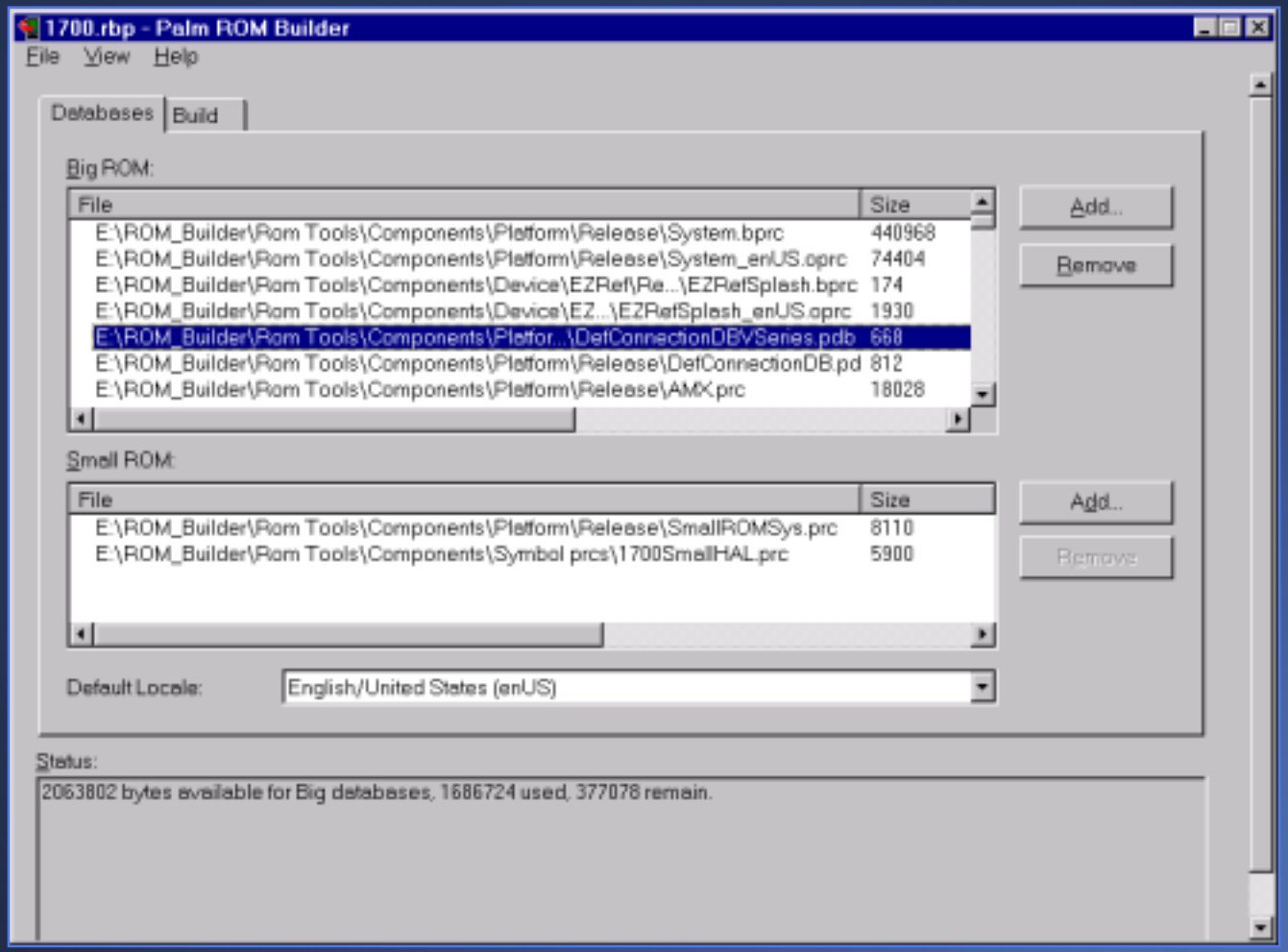
ROM Builder Tool

- Download from the Symbol Website
- Get zip file, i.e., SPTBuild303.zip
- Contains complete 1500 and 17xx projects
- Using ROM Builder for SPT

How to Build a New ROM

- Unzip SPTBuild303.zip
- Launch SymbolROMbuild.exe
- Choose your project
- Open this project, i.e., 1700.rbp
- Add/ Remove files from the ROM image
- Build

How to Build a New ROM Image



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SPT SDK

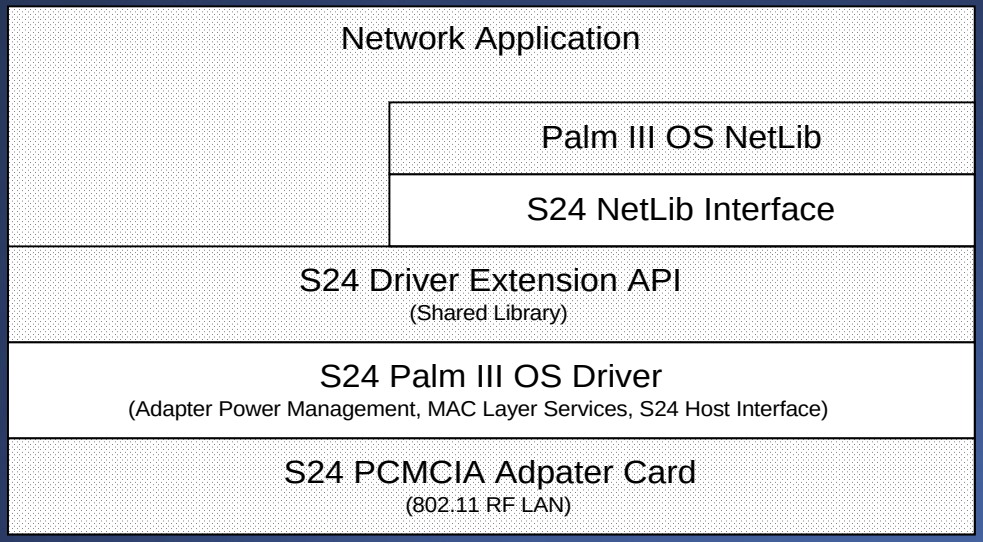
- Scanning
- Printing
- Magnetic Stripe Reader
- Spectrum24
- WWAN

SPT 174x Spectrum 24

- The radio and driver are 802.11 compliant
- Driver supports DHCP and ARP
- Receipt of UDP broadcast messages is supported
- Multicast messages are not supported

SPT 174x Spectrum 24 Driver

- Fully integrated NetIF driver for S24 Radio



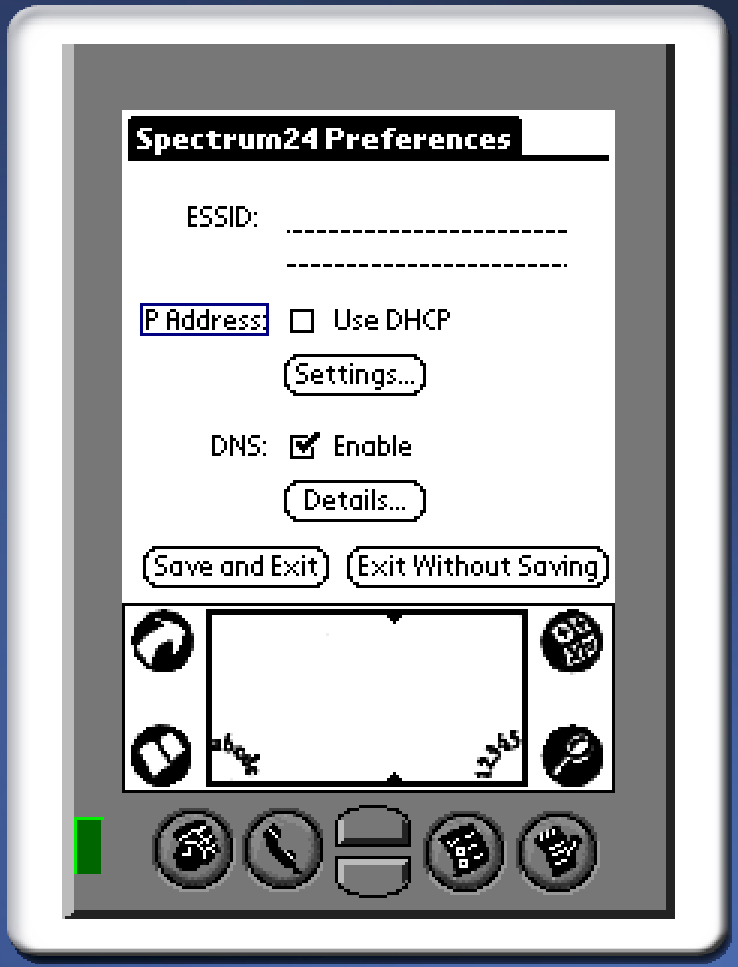
- All applications written for Netlib will function with Spectrum 24
  - Wireless Hot-Sync
  - Scout-Sync
  - AvantGo

SPT 174x Spectrum 24 Setup

- Spectrum 24 is selected from the Network Preference panel service pick-list
- Setup is controlled by pressing the ‘Details’ button on this screen
  - DHCP selection
  - DNS Servers
  - IP address
  - Sub-net Mask
  - Router address

SPT 174x Spectrum 24 Setup

- From the S24 Pref screen you can set
  - ESSID for S24 network
  - IP/DHCP selection
  - DNS enable
- If DHCP is enabled IP, sub-net mask, router, and DNS information is gotten from the server



SPT 174x Spectrum 24 Setup

- When DHCP is disabled you can setup your IP, subnet, and gateway by pressing the settings button



SPT 174x Spectrum 24 Setup

- If DHCP is disabled you will need to setup DNS information if required by your application
- You can specify by name or IP address



SPT 174x Spectrum 24

- The Spectrum 24 driver includes a Extension API for radio specific functions
  - ESSID
  - Network association
  - Driver-specific Preferences
  - Statistics
- Applications are not required to use any of the S24 Extension functions to work properly with the radio

SPT 174x Power Management

- When a Low Battery is detected by the system the radio is shut down, and a S24BatteryError event is generated
- The application should inform the user that this event has occurred, and the net connection should be closed
- Attempting to reopen the connection before the battery has been recharged will cause this sequence to repeat itself

SPT 174x Power Management

- Radio remains on when the unit auto-offs
  - When the unit auto-offs the radio is left on for a period of time, controllable by the application (default is 5 min.) This is know as napping mode. The unit will ‘wake-up’ on receipt of a message
- When a user powers off the unit with the power key it will cause the radio to be powered off immediately

SPT 174x Spectrum 24 SDK

- Symbol SDK includes documentation for the S24 Extensions API
  - Applications are not required to use any of these ‘Side Door’ functions to work properly with the radio
- Sample applications and source code are included
  - Calif DMV app
  - Echo app

Summary—SPT 174x

- SPT 1740
  - 802.11 compliant
  - Radio is a 1 MBPS spread spectrum device
  - FH device
- SPT 1742
  - 802.11 compliant
  - Radio is a 2 MBPS spread spectrum device
  - FH device
- SPT 1746
  - 802.11b compliant
  - Radio is 11 MBPS device
  - Direct Sequencing (DS) device

SPT SDK

- Scanning
- Printing
- Magnetic Stripe Reader
- Spectrum24 Radio
- WWAN

Symbol WAN Terminals

- SPT Models
  - 1733
  - 1734



Two Types of WAN Applications

- Mobile professional
- Commercial/Industrial

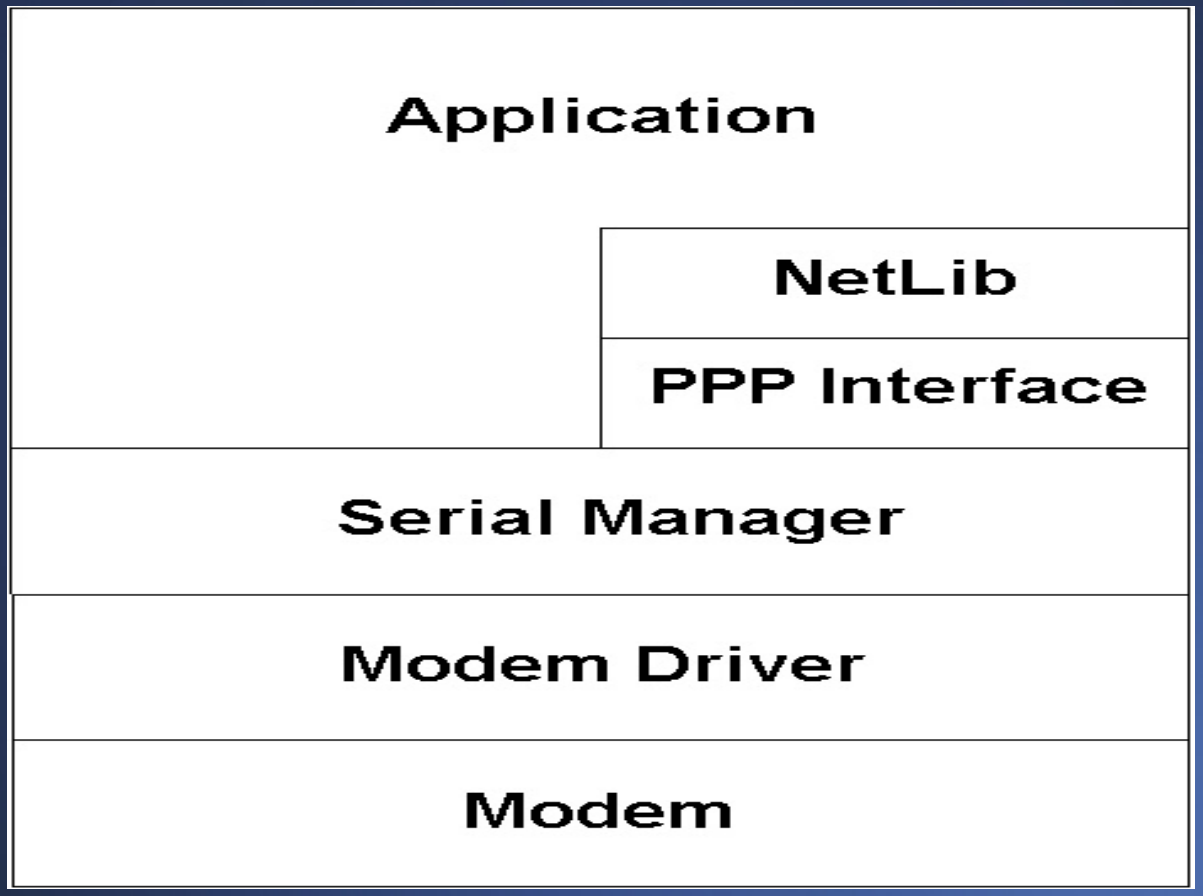
Mobile Professional

- E-Mail
- Messaging/Paging
- Web Browsing
- Characterized by large data volumes

Commercial/Industrial

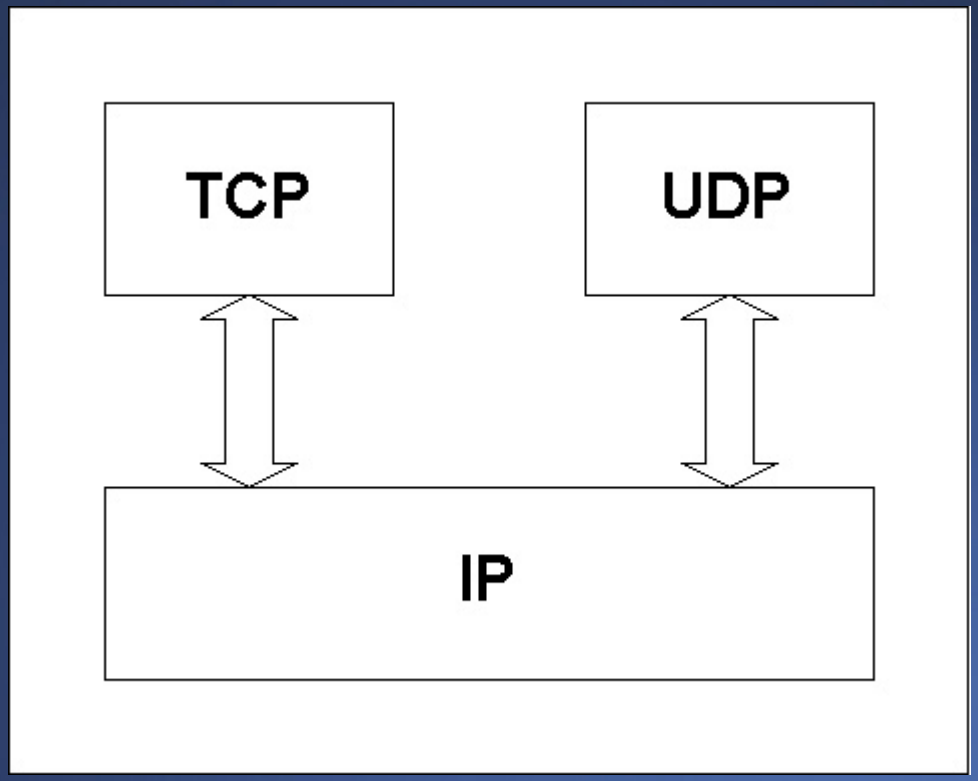
- POS
- Dispatch
- Characterized by short transactions
- Minimum session time
- Vertical solutions

Software Architecture



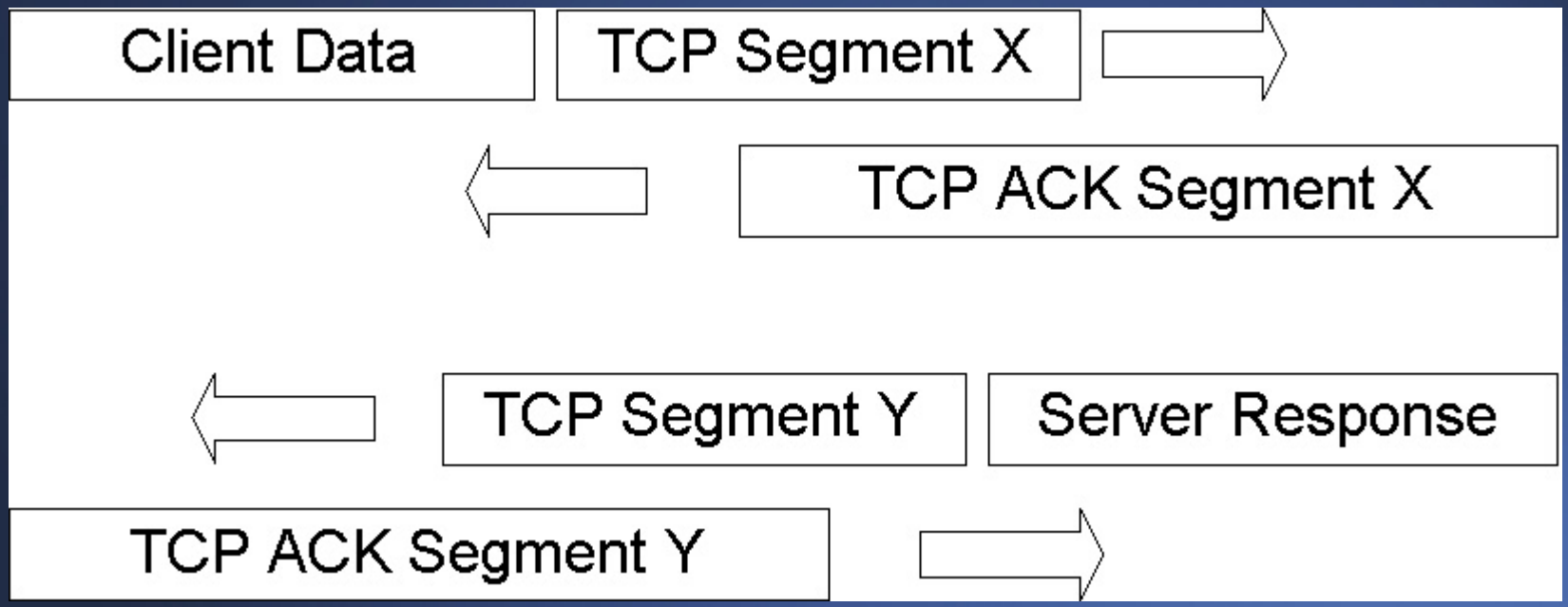
NetLib Protocols

- TCP
- UDP



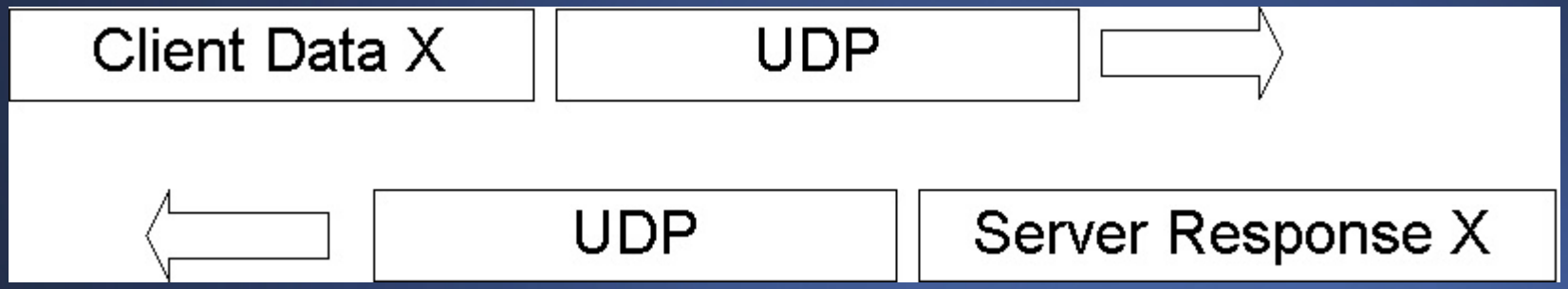
TCP

- Connection oriented
- Reliable



UDP

- Connectionless protocol



Partnering with Symbol

Diana Civelli—Sr. Product Manager,
Software Marketing Business Unit

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Symbol Partner Definitions

- Integrator
- Software Developer
- Value Added Distributor

Symbol's Developer Program

- Symbol Software Partner Program
  - Marketing via Private web site
  - Discounts on demo/development hardware
  - Communications
  - Promotions
  - Business Partner Logo
  - Software Developer's Conference, Trade shows
  - Symbol University
  - Cooperative Marketing Payments

Highlights of Symbol's New Developer Zone

- Zone casts
- Software updates
- Beta testing
- Evaluation copies
- Discounts
- Bulletin boards
- Developer directory
- Related news and events
- Special guest editorials

How to Join the Symbol Software Partner Programs

- Developer Zone
  - Register
    - http://software.symbol.com/devzone
    - Stop by our booth!
- Software Partner Program
  - Register
    - http://www.symbol.com/partner/swdev

Questions and Answers

Thank you for attending this session.

Visit us at the Symbol Technologies Booth in the Solutions Pavilion

174P-179P


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